PROJECT REPORT

Interfacing Four Resistive Sensors

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1 Introduction

Resistive sensors are commonly used for measuring physical quantities such as temperature, pressure, displacement, and light intensity. Most conventional portable sensor systems are designed for only a single sensing channel.

This project focuses on interfacing four resistive sensors using the RC timing measurement technique. In this implementation, four potentiometers are used instead of actual resistive sensors to emulate varying resistance values.

The system uses:

- Arduino Nano
- CD4051 Multiplexer
- Schmitt Trigger
- 100nF Capacitor
- OLED Display

The complete system is battery-powered and portable.

2 Project Objective

The objective of the project is to design a portable multi-channel resistive sensor interfacing system capable of measuring resistance from four channels sequentially.

Main objectives:

- Measure resistance using RC timing technique
- Interface four sensor channels
- Use multiplexing for sensor selection
- Implement Schmitt trigger based threshold detection
- Display readings using OLED display
- Develop a low-cost portable system

3 RC Timing Measurement Principle

The project uses the RC charging principle for resistance measurement.

The capacitor charging equation is:

$$V(t) = V_{cc}\left(1 - e^{-t/RC}\right)$$

Where:

- R = Resistance

- C = Capacitance
- t = Charging time

A fixed capacitor value of 100nF is used in the circuit. The charging time depends on the resistance value.

- Higher resistance → Longer charging time
- Lower resistance → Shorter charging time

The Arduino Nano measures the charging time using the `micros()` function and calculates the resistance.

4 Hardware Components

4.1 Arduino Nano

Arduino Nano controls the complete measurement system. It selects sensor channels, measures charging time, performs calibration, and updates the OLED display.

4.2 CD4051 Multiplexer

The CD4051 multiplexer is used for selecting one sensor channel at a time.

Advantages:

- Reduces hardware complexity
- Reuses a single measurement circuit
- Enables multi-channel sensing

4.3 Schmitt Trigger

The Schmitt trigger converts the analog capacitor voltage into a clean digital signal.

Functions:

- Provides fixed switching threshold
- Improves noise immunity
- Enables accurate digital timing
- Removes the need for ADC conversion

4.4 Potentiometers

Four potentiometers are used to emulate variable resistive sensors.

4.5 100nF Capacitor

A fixed 100nF capacitor is used for RC timing measurement.

4.6 OLED Display

The OLED display shows resistance values of all four channels in real time.

5 System Architecture

The system consists of:

- Four potentiometers
- CD4051 multiplexer
- RC timing circuit
- Schmitt trigger
- Arduino Nano
- OLED display

The multiplexer selects one sensor channel at a time. The capacitor charges through the selected resistance. The Schmitt trigger detects the threshold voltage and sends a digital signal to the Arduino Nano.

The Arduino calculates resistance from charging time and displays the values on the OLED screen.

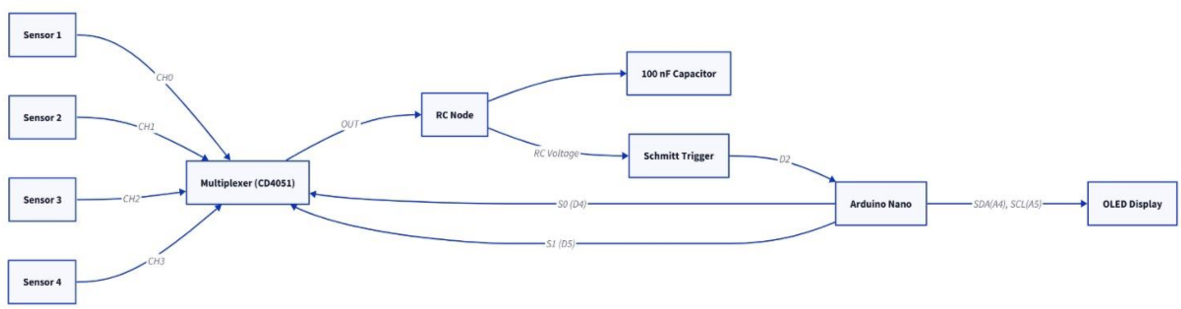

Figure 1: System Architecture

6 Working Principle

The measurement process is carried out as follows:

1. Select sensor channel using multiplexer
2. Discharge the capacitor
3. Start charging capacitor through sensor resistance
4. Detect threshold using Schmitt trigger

5. Measure charging time using `micros()`
6. Convert charging time into resistance
7. Display resistance on OLED display
8. Repeat for next sensor channel

The same RC measurement circuit is reused for all channels through interleaving.

7 Segmented Calibration

To improve accuracy over a wide resistance range, segmented calibration is used.

Segment	Resistance Range
S1	1kΩ – 10kΩ
S2	10kΩ – 20kΩ
S3	20kΩ – 30kΩ
S4	30kΩ – 200kΩ
S5	200kΩ – 500kΩ
S6	500kΩ – 1MΩ

Segmented calibration improves measurement accuracy across the full operating range.

8 Results and Observations

The project successfully demonstrated four-channel resistance measurement using RC timing.

Observations:

- Multiplexer enabled efficient multi-sensor interfacing
- Schmitt trigger improved timing accuracy
- OLED display updated readings properly
- Resistance measurement achieved up to 1MΩ
- Battery operation improved portability

The system operated reliably for all four sensor channels.

9 Image of the Circuit

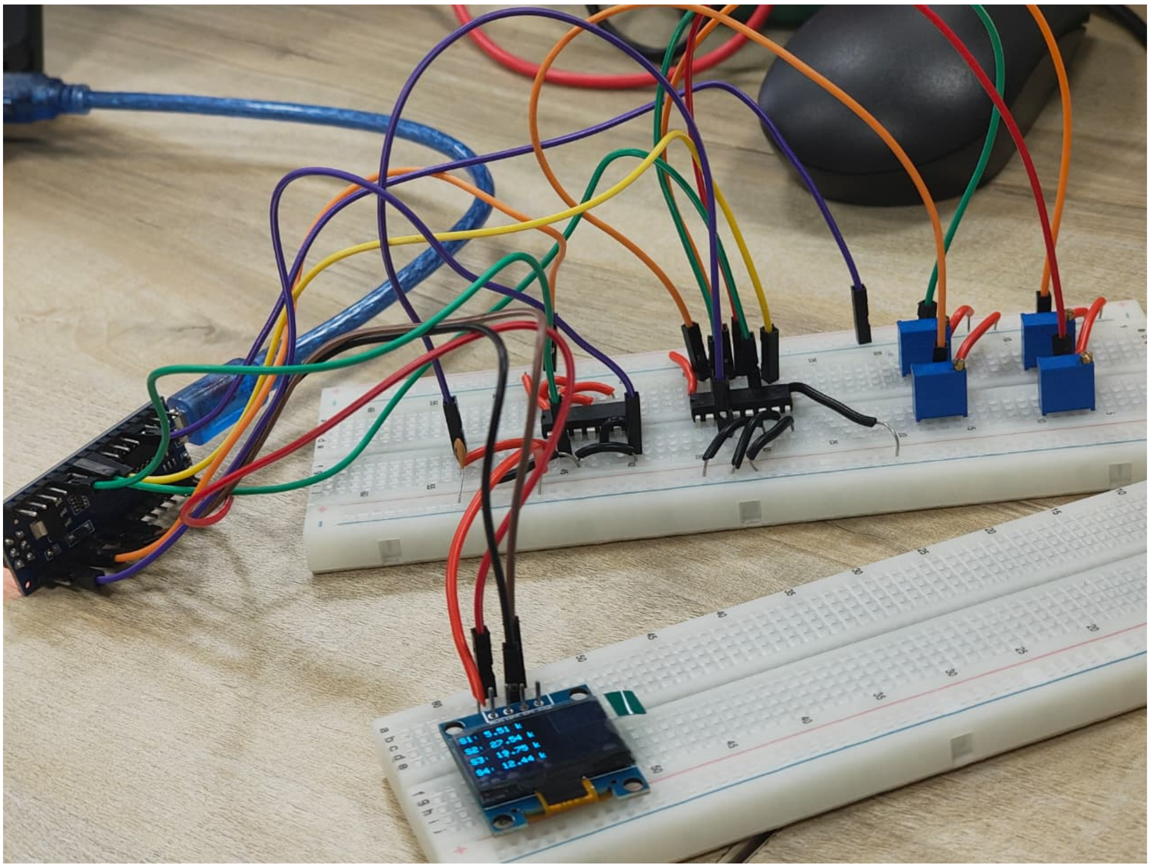

Figure 2: Hardware Implementation of the Project

10 Conclusion

The project successfully implemented a four-channel resistive sensor interfacing system using the RC timing measurement method.

The CD4051 multiplexer enabled efficient sensor selection while sharing a single RC timing circuit. The Schmitt trigger provided reliable threshold detection and improved timing accuracy.

Segmented calibration improved resistance measurement accuracy over a wide range from 1kΩ to 1MΩ.

The project demonstrates a low-cost, portable, and efficient solution for multi-channel resistive sensor interfacing applications.